

BCI based Multi-player 3-D Game Control using EEG for Enhancing Attention and Memory

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Abstract—Brain-Computer Interface (BCI) technique is considered as an efficient alternative modality for improving brain functions such as attention and cognition, based on real time feedback of Electroencephalogram (EEG) signals and their self-regulation. Commercialization of EEG headsets provides tremendous opportunities and possibilities for this technology to employ EEG in video games for cognitive-skill enhancement. This paper proposes a multi-player video game in 3-D environment controlled by EEG features related to 3 different levels of attention. A number of conventional control mechanisms present in commercial games such as keyboard strokes have also been integrated in the game. Three different levels of attention have been detected from players based on their sample entropy features and band power values in alpha, beta and theta bands of EEG. Three subjects have successfully navigated in the designed 3-D environment using EEG based controls as well as keyboard inputs. Experimental results reveal the feasibility of integrating brain signal based inputs along with conventional control inputs in the context of multi-player neurofeedback games for improving brain functions.

Keywords—Neurofeedback, multi-player, attention, Brain-Computer Interface (BCI).

I. INTRODUCTION

Brain-Computer Interface (BCI) is an alternative mode of communication that enables an individual to send commands to a computer or a peripheral device using his brain activity [1]. BCI systems mainly rely on Electroencephalogram (EEG) recordings for measuring brain activity because EEG is one of the most convenient and cheapest brain imaging techniques among the existing non-invasive methods [2]. EEG captures the micro currents produced by the activity of neurons in the brain by placing sensors on scalp. The electrical activity of brain recorded by EEG is susceptible to noise which necessitates the development of robust signal processing algorithms and machine learning techniques for accurate identification of user's intention and successful operation of BCI. Technical advances in engineering and neuroscience has greatly helped EEG based BCI technology to reveal its potential in stroke-rehabilitation and development of neuroprosthetic devices. In addition to its original objective of development of assistive devices for the disabled people [1], BCI research has recently started to focus on multimedia applications such as video games [3]. Video game play has

greatly become a part of entertainment industry, and its positive effect on brain waves has been explored in literature [4].

When BCI technology is employed in video games, it works in a closed loop paradigm where neurofeedback plays an important role. Neurofeedback based systems generally measure brain activity, decode or identify brain patterns of interest, and then provide feedback stimuli to the user depending on the desired state of performance [5, 6]. This real time feedback of EEG signals allows player to self-regulate his specific brain potentials and to gradually rewire the related neuronal networks even for improving certain aspects of brain's attention skills and cognitive power [7]. Neurofeedback training appears particularly promising for individuals diagnosed with attention-deficit hyper active disorder (ADHD) [8]. ADHD is one of the most frequently diagnosed behavioral disorders of children. Worldwide, ADHD is common with an estimated prevalence rate of 5.29%. It is characterized by three behavioral symptoms: inattention, hyperactivity and impulsivity [9]. The primary symptoms of inattention are that either the affected children fail to give close attention easily or they have difficulty in sustaining their attention. Studies comparing neurofeedback to conventional medication reports that neurofeedback as a serious contender for non-pharmaceutical ADHD treatment.

In conventional neurofeedback paradigm, feedback stimuli can be visual, auditory or somatosensory. Programs combining fun and cognitive aspects in neurofeedback training such as BCI based games have great potential for motivating the users and achieving enhancement in brain functions. Training ADHD children even with simple digital console games have been proven as advantageous for improving their brain functions [10]. Games requiring quick and careful selection of control inputs from keyboard help them to co-ordinate sensory information and brain function. Though the game context imposes new challenges due to the complexity of physical and gaming environments when neurofeedback is incorporated in video games, it is worth to exploit its potential for brain function enhancement [7]. It is possible that wide range of movements by a user during gaming may disrupt EEG signals and the game itself. The gaming environment may also disturb the BCI usage as it produces many distractors: visual, tactile or auditory stimuli. In spite of these challenges, video games hold a lot of potential for use in BCIs as they aim to entertain and

motivate the users [11, 12]. A number of studies have already explored the use of EEG-based BCI in a video game context, regarding the interaction techniques and nature of feedback, the performances, or the subjective experience [13]. Recent advances in the brain data acquisition technologies including the emergence of low cost EEG devices [14], make neurofeedback games feasible outside the laboratories too.

In this paper, we focus on a particular interaction paradigm, which is widely used in conventional gaming, but explored by a very few studies in BCI research so far: the multi-user interaction. The objective is to connect multiple users to the same video game application in real-time, through their brain activity. The game is designed such that players have to control their brain activity as well as keyboard inputs timely and effectively to win the game.

The rest of this paper is organized as follows. Section II describes the framework for the proposed system. Gaming interface and its rules are described in Section III. The details of experiments are given in Section IV. Section V analyses and discusses the results of these experiments. Section VI concludes our paper.

II. PROPOSED FRAMEWORK

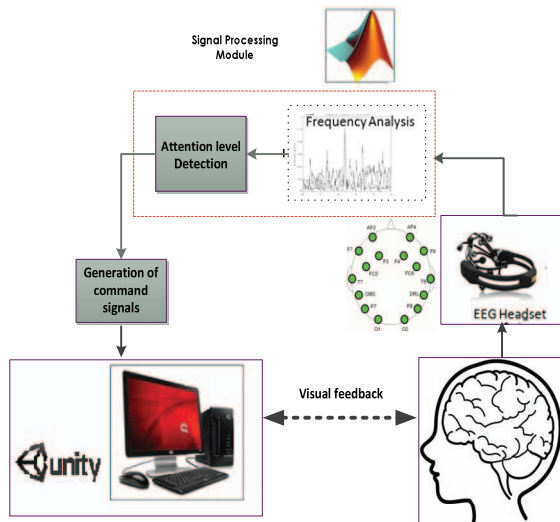


Fig.1 Architecture of the proposed system.

The proposed neurofeedback system consists of a brain signal acquisition unit using Emotiv Epoc neuroheadset, Matlab module for processing EEG signals, control signal generation unit for transforming player's mental state into command signals, the gaming interface designed using C# and unity 3D and visual feedback. The architecture of the system is shown in Fig. 1 and basic modules are briefly explained here.

A. Signal Acquisition Module

The data acquisition is done by Emotiv Epoc neuroheadset which is a low cost EEG recording device comprised of 14 channels of EEG data. The electrodes are located at positions AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8 and AF4 according to 10-20 international system of electrode placement as shown in Fig. 1. The EPOC internally samples at

a frequency of 2048 Hz which is then down-sampled to 128 Hz per channel and the data is then sent to a computer via Bluetooth. It utilizes a proprietary USB dongle to communicate using the 2.4 GHz band. Prior to use, all felt pads on top of the sensors have to be moistened with a saline solution [14].

B. Signal Processing Module

After the data collection stage, the acquired data is transferred to Matlab for further processing. The Matlab module generates attention-related EEG features based on Fast Fourier Transform (FFT) based spectral analysis and sample entropy measurements.

The FFT spectral analysis stage employs 32-point FFT to compute spectral power in theta (4-8 Hz), alpha (8-12 Hz) and beta (13-20 Hz) bands of EEG, as they are the major rhythms in EEG connected to attention and memory [15]. Two parameters, the ratio of alpha power to theta power denoted as α/θ and ratio of beta to alpha power denoted as β/α , are extracted from EEG to assess the attention level of the players. Extensive experimental analysis of EEG spectral power values has been conducted and certain correlations between these values and different levels of attention have been obtained while the player is in an idle state, watching an interesting video clip and performing an IQ test. The α/θ value increases when player begins to concentrate on the video whereas β/α significantly increases when player is focusing and/or actively thinking. Based on these values, player's attention level is classified into 3 levels, Low, Medium and High. Two threshold values are computed to assess these attention levels. The α/θ threshold is the mean of α/θ values of 'idle' state and 'watch video' state. Threshold for β/α is determined by averaging the readings of 'watch video' states and 'answering IQ test' states. If the α/θ value is below α/θ threshold, the attention level is 'Low' (relaxed). If the β/α value exceeds β/α threshold, the attention level is 'High' (concentrated playing). Otherwise, the attention level is considered as medium, assuming the state of playing in a relaxed mind set.

In addition to the proposed combination of band power values, well known sample entropy features are also employed in the neurofeedback system. Sample entropy (*SampEn*) is the rate of information production, and it has been reported in [16] that entropy of EEG signal during attention tasks is greater than that in inattention task. After estimating the *SampEn* features of all EEG channels, the attention score is computed as the average of *SampEn* values obtained from all the selected channels. The procedure for computing *SampEn* is explained in [16]. If the computed attention score is greater than a subject-specific threshold, player's attention is high, otherwise attention is low. The attention levels detected from EEG analysis are then used to generate respective command signals in game. The mapping of EEG features into game control commands and gaming interface are discussed in Section III.

III. GAMING INTERFACE

The main graphical user interface (GUI) of proposed game named as "Mind Battlefield" is shown in Fig. 2. It mainly

consists of the player controlled game character, monsters, other competitors and the battlefield environment. The avatar standing at the middle of GUI shown in Fig. 2 represents the player. Player's ultimate aim is to kill all the monsters and other players found in gaming environment within a given time frame of 180 sec, employing his attention, memory and relevant keyboard inputs. In a holistic view, the game is designed such that player's performance and game score is directly proportional to their cognitive skills such as attention and memory.

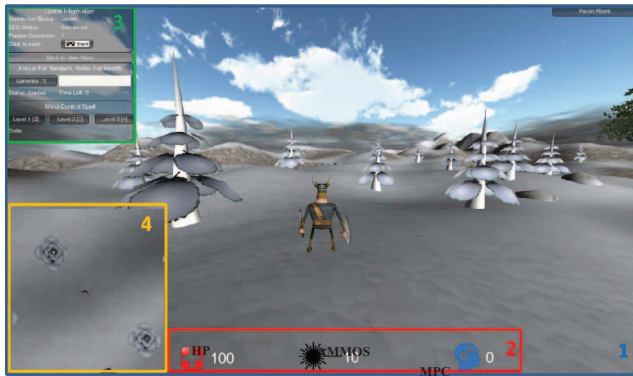


Fig.2 Gaming environment of "Mind Battlefield".

TABLE I. MAPPING OF KEYBOARD CONTROL IN GAME GUI

Key	Action mapped in GUI
Up	Move forward
Down	Move backward
Left	Turn left
Right	Turn right
Space	Jump
Right Mouse Button	Change the View Angle
Middle Mouse Button	Shoot the Laser
Mouse Wheel	Zoom In/Zoom Out

The GUI shown in Fig.2 is divided into 4 main areas. The area marked as 1 (Area-1) represents the main play area for the player controlled character, monsters and other competitors in the battlefield environment. The player's avatar walks in the battlefield with a speed proportional to his attention assessed by entropy analysis. As attention increases, the speed of character also increases. The direction attributes of the player's avatar is controlled using keyboard/mouse and the mapping of keyboard inputs to GUI is given in Table I. The status of player in terms of health points (HP), ammunitions (AMMOS) to kill the monsters and mind control spell (MPC) is displayed in Area-2 of Fig. 2.

In Area-2, the left cross icon represent the HP. The HP of the player is initialized as 100 at the beginning of game. Middle icon indicates the number of AMMOS that the player is carrying. The AMMO is initialized as zero at the beginning of a game. With the AMMOS points, players will be able to attack other players or monsters in the game environment. In order to acquire AMMOS, players have to click on the button, "Generate [1]" as seen in Area-3 in Fig. 1. By doing so, a 15 seconds timer would be triggered and player has to actively concentrate at GUI. At the end of 15 seconds, his attention level would be classified as either "Relaxed", "Low Attention"

or "High Attention" based on α/θ and β/α values. If the player manages to attain the high attention level, AMMOS will be acquired and its value depends on attention level. On the other hand, if they manage to attain the level of "Relaxed", HP would be replenished based on the amount they are relaxed. If the attention score is low then neither AMMOS points nor health points will be achieved.

If the AMMOS runs out, player won't be able to shoot and kill monster or other players. In that case, player can cast the Mind Control Spell to gain control over the monster. It is to integrate a memory control element in the proposed game. The last icon on Area-2 of Fig. 2 is the brain icon which shows the number of Mind Control Spell the player can use. The mind controlled monster will become bigger and go to kill other monster and players automatically. Fig. 3 shows the contrast of the mind controlled monster and normal monster. The bigger monster in green box is the mind controlled one and it will help the player to attack the normal monster which is marked in red box. In order to gain mind control spell, players have to memorize and refill a code of 7, 8, or 9 digits depending on the chosen difficulty levels 1, 2 and 3 respectively. If the player could correctly enter the code within allowed time span, he will get 1, 2 and 3 chances to kill monster for difficulty levels 1, 2 and 3 respectively.

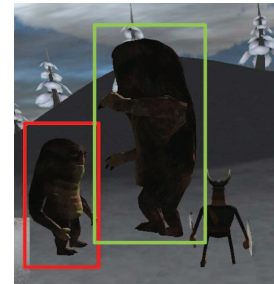


Fig. 3 Mind controlled monster vs. normal monster.

The 4th area in Fig. 1 is the mini-map which is purely a top-down view display screen. It helps the player to monitor the surrounding situation without changing the heading direction, especially when the monster or other player chases him. Fig. 4 shows the snapshot of the interface in competition mode. During this, GUI will display an aiming mark at the middle of the screen to help the player to aim the monster. The green box indicated in Fig. 9 shows this aiming mark towards a monster. Red box indicates the remaining time for current round of game. Game ends when all monsters are killed or when the timer resets to zero.



Fig. 4 GUI during competition mode.

If a player character is killed during game, it will be transported to another game environment with no monsters. Then, player has to attain an attention level “Relaxed”. With that, player’s health will be replenished and transported back to the initial game environment, allowing him to continue with the quest of killing enemies.

The multiplayer feature is implemented with the help of Photon Network application programming interface (API) as shown in Fig. 5. In multi-player model, this API will create a virtual room to keep all the players to be connected together. Players are automatically connected to the same room, “Mind Battlefield”, upon game initialization. Subsequently, upon joining the room, the player’s object will be launched over the network and will be given full control over his own player character. Whenever a player gains a point, a message is sent to Photon Server to update the centralized information stored at the Master Client. The Master Client stores and updates the centralized score effectively. The master client will then periodically send messages using Photon View to broadcast timer information to all clients, ensuring synchronization among all players. As the game round timer is set to be 180 seconds, winner will be decided based on the scores (or number of kills) attained by each player at the end of 180 seconds.

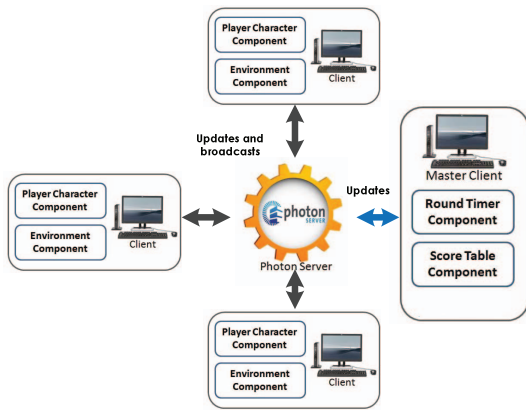


Fig.5 Multi-player network model.

IV. EXPERIMENTAL SETUP

During the experiments, subjects comfortably sit in an armchair facing the computer monitor and wearing the Emotiv Epoc neuroheadset. Signals from 4 EEG electrodes namely O1, O2, AF3, and AF4 are recorded and processed for attention estimation. Players are advised to refrain from unnecessary eye/muscle movements during the experiments for accurate reading.

Before playing the game, every player has to finish a 2-stage training process to estimate threshold values related to α/θ and β/α , and sample entropy. For determining threshold values for α/θ and β/α , subjects are requested to repeatedly perform 3 tasks: simply look at one video clip (low attention task), close their eyes (relaxation task) and answer IQ questions seriously (high attention task). The threshold values of α/θ and β/α are then computed according to the procedure explained in

Section II.B. Then, to derive the individualized threshold value for sample entropy, subject is requested to focus on a ‘plus’ sign displayed on computer screen for 10 sec, and then to refrain from focusing for another 10 sec. Average of these values is taken as threshold for sample entropy based attention detection.

After finishing the training process, the subject can start playing the proposed game which is a multi-player online game and can simultaneously support upto 8 players. Each player compete each other to gain as may enemy kill as possible. The experiment has been conducted on 3 subjects in a quiet environment using 2 work stations. No subject has prior experience in BCI application, but has some experience with first person shooting gaming. Before the experiment begins, they were given some time to play with the game without the BCI elements to familiarize with the control, gameplay and objectives of the game. All 3 subjects managed to pick up basic controls of the game because the keyboard controls used in the game are similar to commercial first person shooting games.

V. RESULTS

Three healthy subjects have played the game for 3 rounds, killing around 16 monsters during each round.

The experimental results explicitly show player’s performance enhancement over time. The impact of the proposed neurofeedback game is evaluated with the following criteria:

- Percentage accuracy in employing desired game control
- Ability of the player to sustain his attention above threshold.

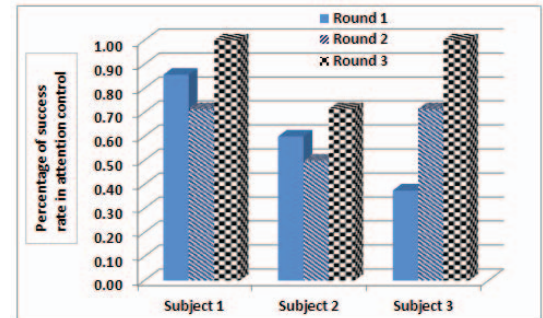


Fig. 6 Success rate in employing desired control strategy.

Before the actual game play, each subject has been given the option to choose the desired skill (EEG features based on concentrate or relax) in order to navigate in the 3-D gaming environment by killing more monsters and keeping himself safe from attacks. Every attempt of the player is noted and if he is able to employ his desired skill as planned, it is considered as a successful attempt. Otherwise it is considered as a failure of player’s control. Success rate is the percentage of successful attempts using the desired skill done by the player during the game. The overall success rates of all players for 3 sessions have been recorded and plotted in Fig. 6. All the 3 subjects improved their attention control skill over time. Graphical analysis of results shows that the proposed BCI game can help players improve their attention level control skills.

In order to measure the ability of a player to sustain his attention level in desired stage, percentage of the time during which player can maintain his attention at the desired attention state during 180 seconds of game play has been computed and shown in Fig. 7. It can be noted that sustainability increases over rounds for every subject, proving the benefits of the proposed neurofeedback game.

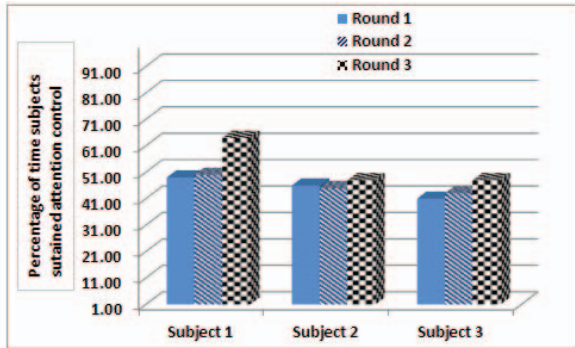


Fig. 7 Desired attention level sustainability rate.

We also present a correlation analysis between the existing sample entropy based attention detection and the proposed attention level detection strategy using α/θ and β/α values. Figures 8 and 9 show the attention levels detected using α/θ and β/α values, and attention score estimated using sample entropy values when Subject-1 concentrates and relaxes for a period of 15 seconds respectively. It can be found that whenever the attention score computed by entropy is greater than the threshold, attention level is identified as high as in Fig. 8 whereas that whenever the attention score computed by entropy is lower than the threshold, attention level is identified as low as in Fig. 9. The graphs show clear correlation between the 2 parameters used in our neurofeedback game. This trend has been observed for all the subjects analyzed. Though an obvious relationship exists between both strategies, further investigation is necessary to conclude the superiority of one technique compared to the other.

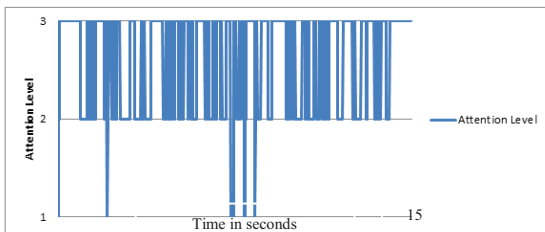


Fig. 8 (a) Detected attention levels using α/θ and β/α during concentration.

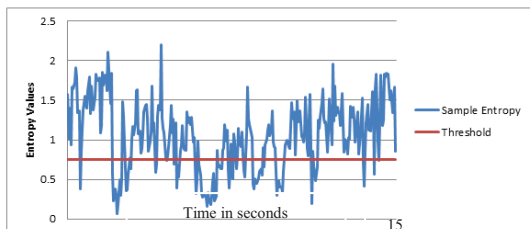


Fig. 8 (b) Sample entropy values during concentration.

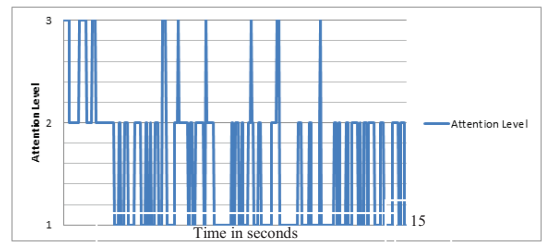


Fig. 9 (a) Detected attention levels using α/θ and β/α during relaxation.

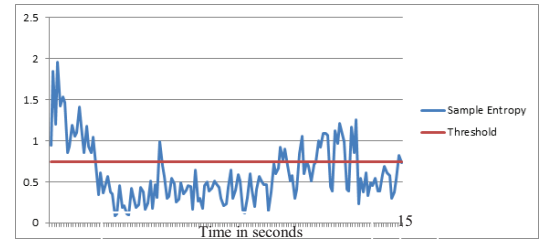


Fig. 9(b) Sample entropy values during relaxation.

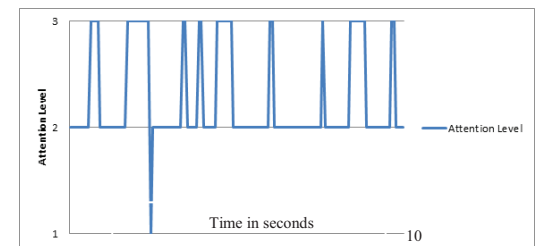


Fig. 10 (a) Detected attention levels using α/θ and β/α .

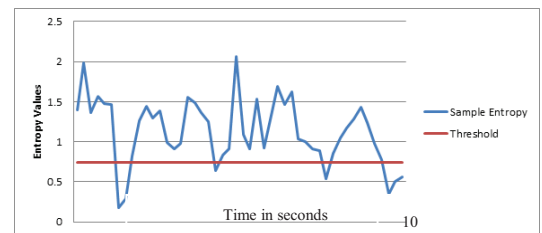


Fig. 10 (b) Sample entropy values during memorization.

The attention level and sample entropy values during a span of 10 seconds when players are given a string of numbers to memorize during the game are also plotted in Fig. 10. In Fig. 10(a), it is found that attention level is at level 3 (High attention state) for most of the time. The majority of entropy values are also above the threshold as in Fig. 10(b). It is interesting to note the high attention state during memorization task. This could be due to the high visual focus during the span of 10 seconds when they are supposed to remember the string of numbers.

The proposed BCI system successfully generated and utilized control inputs directly from brain activity, along with conventional control inputs from keyboard for playing a 3-D video game. The system effectively monitors the attention of players using entropy and band power values of EEG throughout the entire game. It has been found that the proposed control mechanism in the designed video game is capable of enhancing attention and brain functions.

VI. CONCLUSION

This paper proposed a 3-D video game driven by EEG features related to different levels of attention and a set of keyboard inputs. The game named as “Mind Battle Field” employs sample entropy as well as band power estimates of alpha, beta and theta rhythms of EEG to differentiate between different brain states of players. The complex attention control mechanism proposed in this paper helps players to improve attention and over all game control skills. Game scores obtained for 3 subjects increases over 3 rounds of game play and it explicitly shows the effect of learning mechanism on player’s brain functions and control actions. Experimental results show the promising capability of the neurofeedback in 3-D game environment for enhancing game performance. Further experimental analysis is necessary to make the control mechanism simpler and more robust for optimizing the benefits of neurofeedback training.

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